Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. CSE	**Semester**	VI
Subject Code	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	**Batch**	2023–2027
Name	**Jaswanth Sridharan**	**Reg. No.**	**3122235001059**
Due Date	20.01.2026		

Aim

To explore and work with Python packages like Numpy, Scikit-learn, and Matplotlib on datasets from public repositories and identify ML tasks, feature selection techniques, and suitable algorithms.

Libraries Used

- Numpy (imported as `np`)
- Pandas (imported as `pd`)
- Matplotlib.pyplot (imported as `plt`)
- Seaborn (imported as `sns`)
- OpenCV (`cv2`)
- OS (Standard Library)
- Math (Standard Library)

Mathematical and Theoretical Description

1. **Exploratory Data Analysis (EDA):** Summarize distributions, detect outliers, assess class balance, and reveal pairwise relationships/correlations.

2. **Descriptive Statistics:**

   - Mean: $\mu = \frac{1}{n}\sum x_i$
   - Sample variance: $s^2 = \frac{1}{n-1}\sum(x_i - \mu)^2$
   - Quantiles (Median, $Q1$, $Q3$) used for spread and boxplot construction.

3. **Histograms:**

   - Empirical density approximation via bin counts.

- Kernel Density Estimate (KDE):

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

  where K is the kernel (e.g., Gaussian) and h is the bandwidth.

4. Boxplots:

- Shows Median, $Q1$, $Q3$; Interquartile Range $IQR = Q3 - Q1$.
- Whiskers extend to min/max within $[Q1 - 1.5 \cdot IQR,\ Q3 + 1.5 \cdot IQR]$.

5. Correlation Heatmap:

- Pearson correlation coefficient:

$$r_{XY} = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}, \quad \mathrm{cov}(X, Y) = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$$

Results and Discussions

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2	Img/img001-002.png	0
3	Img/img001-003.png	0
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3407	Img/img062-052.png	z
3408	Img/img062-053.png	z
3409	Img/img062-054.png	z
3410	Img/img062-055.png	z

3411 rows × 2 columns

Figure 1: Dataset Overview

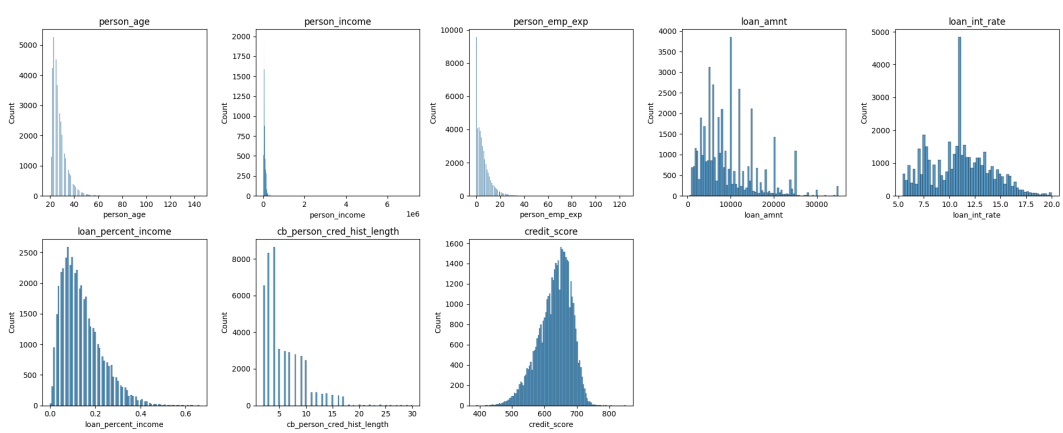

Figure 2: Data Visualization

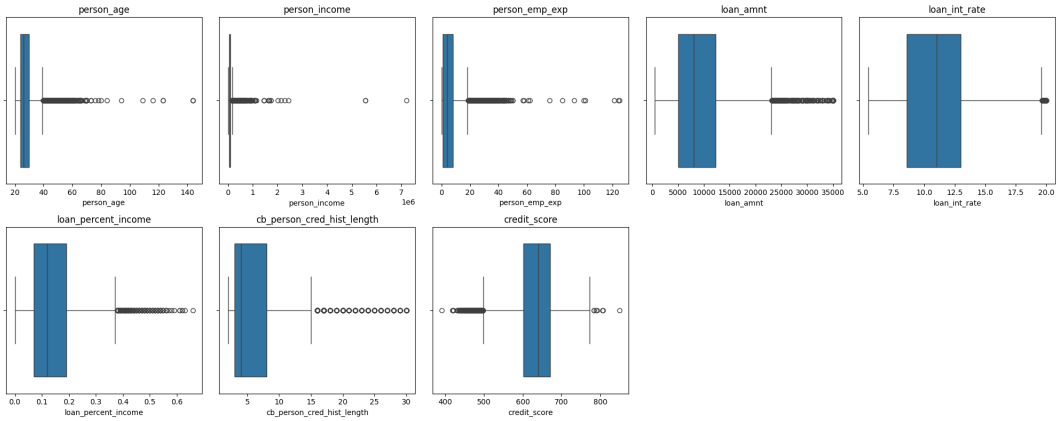

Figure 3: Feature Distribution

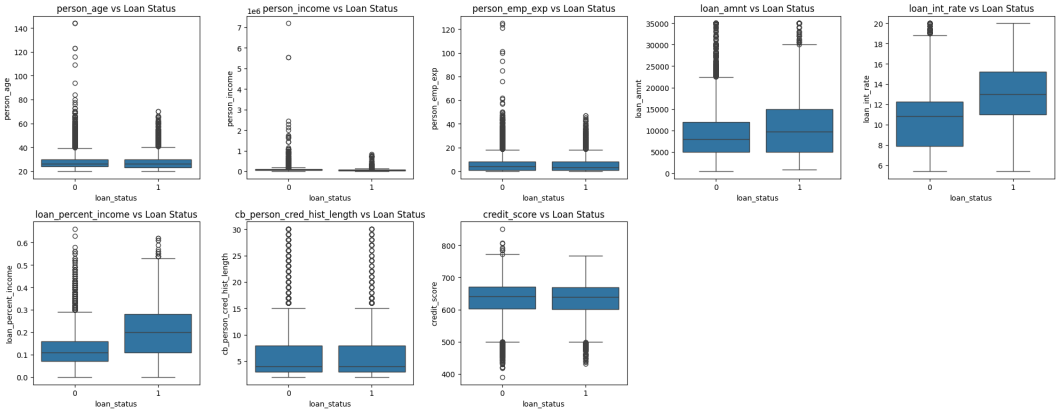

Figure 4: Feature Relationships

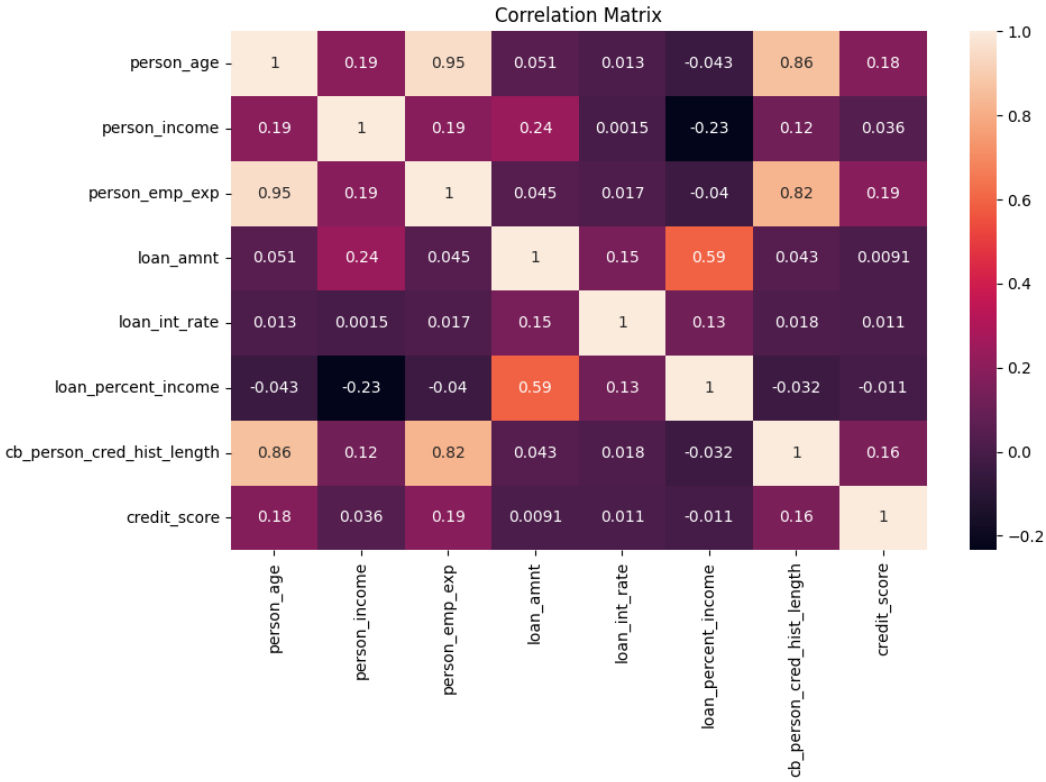

Figure 5: Correlation Matrix

Task Identification Table:

Dataset	ML Task	Feature Selection	Suitable Algorithm
Iris Dataset	Multi-class Classification	Correlation Matrix / ANOVA F-value	k-NN, Decision Trees
Loan Amount	Regression	Recursive Feature Elimination (RFE)	Linear Regression, Random Forest
Diabetes	Binary Classification	Chi-Square Test / SelectKBest	Logistic Regression, SVM
Email Spam	Binary NLP	Information Gain	Naive Bayes, SVM
MNIST	Image Classification	PCA	CNN, SVM

Learning Practices

- **Interpret dataset structure:** Learn to inspect shape, info, and missing values.

- **Visualize distributions:** Gain skills in plotting histograms, boxplots, and heatmaps.
- **Identify class balance:** Understand how label distribution affects model performance.
- **Spot feature relationships:** Use pairplots and correlation matrices.
- **Perform dimensionality reduction:** Use PCA for visualization.